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Course

Evolutionary

Algorithms

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Semester

02

Task

02

Real-Coded Genetic
Algorithm

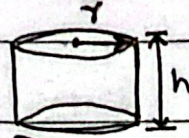
Electrical Department AI

University of Engineering
And Technology Peshawar.

Lecture 04 Task:

Real-coded GA for water Tank Design Optimization.

Task Title:-



Optimize the Dimensions of a Cylindrical Water Tank Using Real-coded Genetic Algorithm.

Problem Statement:-

A company wants to design a closed cylindrical water tank that can store at least

$$\rightarrow V_{\min} = 1 \text{ m}^3$$

The tank has two design variables:

$$\rightarrow r = \text{tank radius}$$

$$\rightarrow h = \text{tank height}$$

The Goal is to find the best values of r & h such that material cost is minimized while satisfying the required volume constraint.

Design variables & Bounds:-

Variables	Meaning	Lower Bound	Upper Bound
r	Tank radius	0.1 m	1.5 m
h	Tank height	0.2 m	4.0 m

So

Each chromosome is a real-valued

Vector:

$$X = [r, h]$$

The volume of the cylinder is:

$$V = \pi r^2 h$$

Tank must satisfy

$$V \geq 1 \text{ m}^3$$

Cost Function:

Assume

Tank is closed from the top & bottom.

Side wall:

$$2\pi r h \times 500$$

Top + Bottom:-

$$2\pi r^2 \times 700$$

Total Cost:

$$C = 500(2\pi r h) + 700(2\pi r^2)$$

$$C = 1000\pi r h + 1400\pi r^2$$

Penalty:-

Because the tank must have at least 1 m^3 volume, use a penalty if the volume is less than 1:

if solution is invalid (volume < 1).
Push it

$$\text{Penalty} = 100000(1-v)$$

if valid:

$$\text{Penalty} = 0$$

Final Objective Function:-

$$F = C + \text{Penalty}$$

This is what we minimize

Theoretical Optimum:

In this step, I try to find the minimum cost using mathematics before applying the Genetic Algorithm.

I assume that the value is exactly $v = 1 \text{ m}^3$.

Because if the volume becomes more than 1,

than 1, the cost will also increase,

So, minimum cost will occur when the volume is equal to the required value.

Volume Formula:-

$$V = \pi r^2 h$$

Put $V = 1$

$$1 = \pi r^2 h$$

Solve for h

$$h = \frac{1}{\pi r^2}$$

We convert the equation into a single variable (r)

Substitute into Cost

Original Cost:

$$C = 1000 \pi r h + 1400 \pi r^2$$

Substitute h :

$$C = 1000 \pi r \left(\frac{1}{\pi r^2} \right) + 1400 \pi r^2$$

$$C(r) = \frac{1000}{r} + 1400 \pi r^2$$

Now the cost depends only on radius r . This allows differentiation to find the minimum value.

Take Derivative:-

$$\frac{dc}{dr} = -\frac{1000}{r^2} + 2800\pi r$$

Set derivative = 0

$$\Rightarrow -\frac{1000}{r^2} + 2800\pi r = 0$$

Take derivative to find The minimum point of the cost function.

Solve for r

$$2800\pi r = \frac{1000}{r^2}$$

$$(*) \quad 2800\pi r^3 = 1000$$

$$r^3 = \frac{1000}{2800\pi}$$

Multiply both sides by r^2 in the Second Step (*)

$$r^3 = \frac{1000}{2800\pi}$$

$$r^3 = 0.1137$$

$r = 0.48$ This gave the optimal value that minimizes the cost.

Find h

$$h = \frac{1}{\pi r^2}$$

$$h = 1$$

$$\Rightarrow h = 1.38$$

$$(3.1416) \times (0.48^2)$$

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$$h = 1.38$$

Now I use the value of r to calculate height.

Final Cost:-

$$C = 1000 \pi r h + 1400 \pi r^2$$

$$C = 3100$$

This is the minimum cost when the volume constraint is satisfied.

variables	value
r	$r = 0.48m$
h	$h = 1.38m$
Cost	$C = 3100$

From this step, I learned that the minimum cost occurs when the values is exactly equal to the required value.

This result will help me later to compare with the Genetic Algorithm results.

Calculation:-

Initial Population:-

In This Step, I randomly select some values of radius (r) & height (h) within the given limits. These values form the Initial Population for the Genetic Algorithm.

Given Bounds:-

$$\rightarrow 0.1 \leq r \leq 1.5$$

$$\rightarrow 0.2 \leq h \leq 4.0$$

Initial Population:-

Index	$r(m)$	$h(m)$
1	0.6	1.2
2	1.3	0.4
3	0.4	2.5
4	0.9	0.7

These values are selected randomly but within the valid range.

Each pair (r, h) represents one possible solution.

Calculate volume, Cost,
Penalty & F

Formula Used:

Volume:

$$V = \pi r^2 h$$

Cost:-

$$C = 1000\pi rh + 1400\pi r^2$$

Penalty:-

$$\text{if } v < 1 \rightarrow \text{Penalty} = 100000(1-v)$$

$$\text{if } v \geq 1 \rightarrow \text{Penalty} = 0$$

Calculations:-

Index 1: (0.6, 1.2)

$$r, h = 0.6, 1.2$$

Putting values in volume formula

$$V = 3.1416 \times (0.6)^2 \times 1.2 = 1.357$$

$$C = 1000 \times 3.1416 \times 0.6 \times 1.2 + 1400 \times 3.1416 \times (0.6)^2$$

$$C = 2261.9 + 1583.4 = 3845.3$$

$$\text{Penalty} = 0$$

$$F = 3845.3$$

Index 2 (1.3, 0.4)

$$v = 1.3$$

$$h = 0.4$$

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$$V = 3.1416 \times (1.3)^2 \times 0.4 = 2.123$$

$$C = 6534.0$$

$$\text{Penalty} = 0$$

$$F = 6534.0$$

Index 3: (0.4, 2.5)

$$V = 3.1416 \times (0.4)^2 \times 2.5 = 1.256$$

$$C = 3.141.6 + 703.7 = 3845.3$$

$$\text{Penalty} = 0$$

$$F = 3845.3$$

Index 4: (0.9, 0.7)

$$V = 3.1416 \times (0.9)^2 \times 0.7 = 1.781$$

$$C = 1979.2 + 3564.2 = 5543.4$$

$$\text{Penalty} = 0$$

$$F = 5543.4$$

Index	r	h	Volume	Cost	Penalty	F
1	0.6	1.2	1.357	3845.3	0	3845.3
2	1.3	0.4	2.123	6534.0	0	6534.0
3	0.4	2.5	1.256	3845.3	0	3845.3
4	0.9	0.7	1.781	5543.4	0	5543.4

In this step, I generated random solutions & calculated their cost & feasibility. Since all solutions satisfy the volume constraint, no penalty is applied.

Tournament Selection:

In this step, I select the best individuals from the population.

I compare two solutions at a time & choose the one with lower objective value (F).

Rule:-

Lower F value = Better solution.

Initial F values:-

Index	r	h	F
1	0.6	1.2	3845.3
2	1.3	0.4	6534.0
3	0.4	2.5	3845.3
4	0.9	0.7	5543.4

Tournament Matches:

Match 1:

Index 1 (3845.3) vs Index 2 (6534.0)

Winner → Index 1

Match 2:-

Index 3 (3845.3) vs Index 4 (5543.4)

Winner Index 3

Match 3:-

Index 2 (6534.0) vs Index 4 (5543.4)

Winner Index 4

Match 4:-

Index 1 (3845.3) vs Index 3 (3845.3)

Winner → Index 1

Mating Pool

Index	r	h	F
1	0.6	1.2	3845.3
3	0.4	2.5	3845.3
4	0.9	0.7	5543.4
1	0.6	1.2	3845.3

In this step, better solutions are selected for the next process. Individuals with lower cost have more chance to be selected. This helps the

Algorithm move better solutions.

SBX Crossover:

In this step, I combine two parent values to create new children.

The children are almost similar to parents but slightly different.

Selected Parents:-

Pair 1

$$P_1 = (0.6, 1.2)$$

$$P_2 = (0.4, 2.5)$$

Pair 2

$$P_3 = (0.9, 0.7)$$

$$P_4 = (0.6, 1.2)$$

Assume values:

$$U = 0.7$$

$$\beta = 1.02$$

β is called spread factor.

It controls how close the children are to parents.

Now finding Children Pair 1

Child 1:

$$v = (0.6 + 0.4) / 2 = 0.5$$

$$\rightarrow \text{adjust} = 0.61$$

First take average, then
Slightly adjust due
to crossover effect.

$$h = (1.2 + 2.5) / 2 = 1.85$$

$$\rightarrow \text{adjust} = 0.61$$

Same method applied for height.

So

$$C_1 = (0.61, 1.17)$$

Child 2:-

$$v = 0.39$$

$$h = 2.53$$

This child is formed using
the opposite combination

So

$$C_2 = (0.39, 2.53)$$

Now Pair 2

Child 3:

$$v = 0.91$$

$$h = 0.69$$

Same Crossover Process is repeated.

So

$$C_3 = (0.91, 0.69)$$

child 4:

$$r = 0.59$$

$$h = 1.21$$

values remain close to parents.

So

$$C_4 = (0.59, 1.21)$$

Final Children Before Mutation

Child	r	h
C ₁	0.61	1.17
C ₂	0.39	2.53
C ₃	0.91	0.69
C ₄	0.59	1.21

In this step, I created new solutions by combining parent values.

The children are similar but not exactly the same, which helps in exploring better solutions.

Polynomial Mutation:-

In This step, I Slightly change The children values.

This helps to ~~any~~ Introduce variation and avoid getting stuck at one solution.

Children Before Mutation.

$$C_1 = (0.61, 1.17)$$

$$C_2 = (0.39, 2.53)$$

$$C_3 = (0.91, 0.69)$$

$$C_4 = (0.59, 1.21)$$

Apply Mutation:-

$$C_1 (0.61, 1.17) \rightarrow (0.60, 1.16)$$

Slight decrease in both values.

$$C_2 (0.39, 2.53) \rightarrow (0.40, 2.55)$$

Slight increase

$$C_3 (0.91, 0.69) \rightarrow \text{0.91, 0.69}$$

$$\rightarrow (0.92, 0.68)$$

Small adjustment

$$C_4 (0.59, 1.21) \rightarrow (0.58, 1.22)$$

Minor variation applied

Children After Mutation:-

child	r	h
C_1	0.60	1.16
C_2	0.40	2.55
C_3	0.92	0.68
C_4	0.58	1.22

Mutation slightly changes the values of r & h . It helps maintain diversity in the population & improves the chances of finding a better solution.

Final Evaluation / Initial + offspring.

Final Evaluation & Next Generation Selection.

In this step, I combine initial population & mutated children. Then I calculate volume, cost, Penalty &

Penalty & Objective (F) for each, & The best Solutions.

Formula used:-

Volume:

$$V = \pi r^2 h$$

Cost

$$C = 1000\pi r h + 1400\pi r^2$$

Penalty:

$$\text{if } V < 1 \rightarrow \text{Penalty} = 100000(1-V)$$

$$\text{Else} \rightarrow \text{Penalty} = 0$$

Objective:

$$F = C + \text{Penalty}$$

All Population (Initial + children)

Index	r	h	V	Cost	Penalty	F
P ₁	0.6	1.2	1.357	3770	0	3770
P ₂	1.3	0.4	2.123	6760	0	6760
P ₃	0.4	2.5	1.256	3260	0	3260
P ₄	0.9	0.7	1.780	5230	0	5230
C ₁	0.60	1.16	1.313	3680	0	3680
C ₂	0.40	2.55	1.282	3300	0	3300
C ₃	0.92	0.68	1.810	5350	0	5350
C ₄	0.58	1.22	1.287	3720	0	3720

All values are calculated using formulas.

Sort by objective (F)

Rank	Solution	γ	h	F
1	P_3	0.4	2.5	3260
2	C_2	0.40	2.55	3300
3	C_1	0.60	1.16	3680
4	C_4	0.58	1.22	3720

Next Generation

$$P_3 = (0.4, 2.5)$$

$$C_2 = (0.40, 2.55)$$

$$C_1 = (0.60, 1.16)$$

$$C_4 = (0.58, 1.22)$$

These solutions are selected because they have the lowest objective (F) values among all individuals.

In Genetic Algorithm, lower F means cost, which is better.

I arranged all solutions in ascending order of F and selected the top 4. These act as parents for the next generation. Also all

Selected Solutions Satisfy The
Volume Constraint ($V \geq 1$)

So no penalty is applied.

Questions To Answer:

Q 1. Why is real-coded GA
Suitable for this problem?

Answer: Real-coded ~~is~~ - real-valued
Genetic Algorithm is a good
choice because the design
variables (radius & height)
are continuous in nature.
It works directly with
decimal values, which makes
the solution more precise
& avoids unnecessary
conversions.

Q 2. What is the effect if the
calculated volume is below
 1 m^3 ?

Answer. If the volume goes below
 1 m^3 , the solution does
not meet the requirement.

requirement. In that case a penalty is added to increase the objective value, making the solution less favorable during solution.

Q 3. What is the purpose of introducing a penalty in the objective function?

Answer The purpose of the penalty is to enforce the constraint it discourages solutions that do not satisfy the minimum volume condition by increasing their overall cost.

Q 4. How does SBX crossover contribute in the optimization problem?

Answer SBX crossover generates new candidate solutions by combining two parent solutions. It allows the algorithm to explore different combinations of radius & height, which helps in finding improved results.

Q 5. why is mutation applied after crossover?

Answer- Mutation is applied to introduce small random changes in the solution. This helps maintain diversity in the population & reduces the chance of the algorithm getting stuck in local optimum.

Q 6. Do the final selected solutions meet the volume requirements?

Answer Yes the final selected solutions satisfy the condition of minimum volume ($V \geq 1m^3$),

so no penalty is applied to them.

Bonus Question:-

Does the GA always provide the same result in every run?

Ans No, the results may vary in each run.

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each run. The GA
Uses randomness in Selection,
Crossover, & mutation.
Different random values
can lead to slightly
different solutions.
